

# Do Markets Price Climate Disclosure? A Large Language Model Approach

Haiping Wang and Xin Zhou(✉)

Volatility Institute, New York University Shanghai, Shanghai, China
{hw2942,xinzhou}@nyu.edu

**Abstract.** We employ machine learning (ML), deep learning (DL), and fine-tuned large language models (LLMs) to construct climate variables from Corporate Social Responsibility (CSR) disclosures, ESG reports, and climate risk alerts of A-share listed firms in China. Our results show that LLMs consistently outperform traditional ML and DL approaches across all tasks, achieving the highest accuracy and F1 scores. Support Vector Machines (SVM) also deliver competitive performance in simpler classification settings, indicating their potential as a cost-effective alternative in resource-constrained environments. We further examine the relationship between LLM-generated climate variables and stock returns, revealing that while climate opportunity, climate and transition risk are priced by markets, overall climate-related content and physical risk are not. These findings suggest that China's equity market selectively prices climate information.

**Keywords:** Machine Learning · Deep Learning · Large Language Models · Climate Text Classification · Climate Risk

## 1 Introduction

The increasing frequency and severity of extreme weather events, from heat-induced droughts to catastrophic floods, highlight the urgency of addressing climate change. This has intensified public attention and underscored the need for accurate monitoring of climate-related information, including meteorological data, public awareness, policy developments, and corporate actions. Reliable measurement enables regulators to design effective policies and helps investors manage climate risks. Beyond numerical indicators such as temperature and precipitation, researchers now increasingly exploit text-based sources—news, social media, and corporate reports. Advances in data availability, computing power, and algorithms have driven climate text analysis from dictionary and bag-of-words methods to modern ML, DL, and LLM approaches.

ClimateBERT [23], a transformer-based language model pre-trained on over two million climate-related text paragraphs from news articles, research abstracts, and corporate climate reports, demonstrates superior performance

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over models like ChatGPT in climate-related text classification tasks [21]. Applications of ClimateBERT include classification, sentiment analysis, and fact-checking for climate-related texts. Further extensions have identified corporate climate commitments, assessed the specificity of climate statements [3], and analyzed climate-risk disclosures within the Task Force on Climate-related Financial Disclosures (TCFD) framework [2]. A specialized variant, ClimateBERT-NetZero [19], has been trained to detect texts referencing net-zero or reduction targets, with additional applications in conventional question-answering models and corporate earnings call analyses. Moreover, BERT-based models have quantified regulatory climate risk disclosures into physical and transition risk categories [15], contributing to a deeper understanding of climate-related impacts and solutions.

Advancements in Chinese climate-related LLMs are providing new opportunities to address environmental concerns within China's linguistic and cultural context. EnvBERT [1], a Bidirectional Encoder Representations from Transformers (BERT) model pre-trained on a 13.5 GB environmental corpus, serves as a foundational model for EnvText, an accessible Natural Language Processing (NLP) software package designed for mining and analyzing Chinese environmental text. The Chinese Climate Policy Uncertainty (CCPU) index [18] has been constructed by fine-tuning MacBERT-Base-Chinese on a corpus of six newspapers with manual auditing, marking the first development of multi-level CCPU indices at national, provincial, and city scales with daily, monthly, and annual frequencies. Similarly, climate action disclosures have been identified using a fine-tuned BERT-Base-Chinese model applied to Q&A transcripts from earnings calls of A-share listed companies in China [20]. To explore the narrative framing of climate change values across cultural landscapes [24], GPT-4 has been employed to analyze climate-related news articles from North American and Chinese media, offering comparative insights. While English LLMs have achieved widespread application across various domains, Chinese climate-related LLMs show substantial potential for further refinement, particularly in achieving more detailed and precise classification and application capabilities.

To address the gap in Chinese climate-related LLMs, this study enriches the field across three dimensions: data, models, and applications. We collect diverse datasets, including CSR and ESG reports from A-share listed companies in China and climate risk alert information from Wind. To evaluate model performance, we train and test ML models, DL models, and LLMs. The ML models include k-nearest neighbors (KNN), Naive Bayes, Logistic Regression, Decision Tree, Random Forest, and SVM. In the DL category, Long Short-Term Memory (LSTM), bidirectional LSTM (BiLSTM), and Gated Recurrent Unit (GRU) are examined. Additionally, for LLMs, basic models such as BERT-Base-Chinese, Chinese-MacBERT-Base, and RoBERTa are fine-tuned to assess their relative performance in comparison to the other model types. We focus on three progressive classification tasks: identifying whether the text is climate-related or not, performing sentiment analysis to distinguish climate risks or opportunities, and categorizing climate risks into physical or transition risks. These tasks are

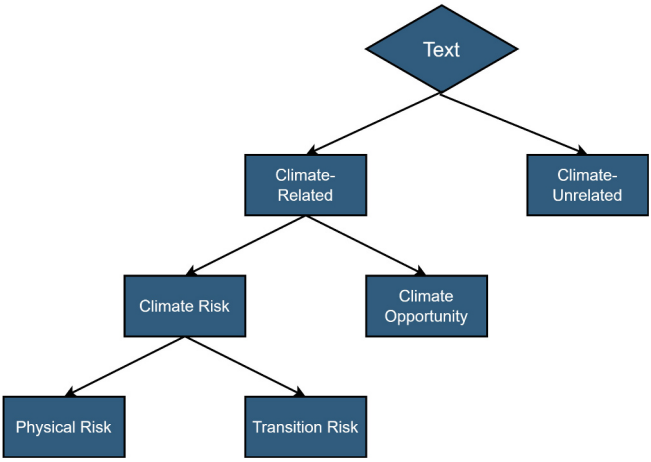



**Fig. 1.** The hierarchical structure of three climate-related classification tasks.

interconnected, forming a hierarchical structure, as illustrated in Fig. 1. Empirical findings illustrate that LLMs continuously outperform ML and DL models in accuracy and F1 scores across all tasks. Among ML models, SVM achieves remarkable performance in simple climate-related classification tasks, making it a viable alternative in scenarios with limited computational resources and training time. For financial applications, we apply a fine-tuned BERT-base-chinese model to extract climate-related content from ESG and CSR reports of A-share listed firms (20072024), constructing five firm-year climate metrics. Linking these to monthly returns reveals that climate risk and transition risk are negatively priced, while climate opportunity is positively priced. In contrast, overall climate disclosure volume and physical risk show no significant return effects.

This study advances Chinese climate-related LLMs in four dimensions. First, we build datasets from diverse sources, including corporate reports and climate risk alerts from Wind, integrating both internal disclosures and external news. Second, we benchmark three fine-tuned LLMs against a broad set of ML and DL models, ensuring robust comparisons. Third, we design three climate-related classification tasks and a structured workflow for data collection, model training, and evaluation, with financial applications demonstrating how textual climate data can inform investment strategies and sustainability reporting. Finally, we release all labeled datasets and fine-tuned LLMs on Hugging Face (https://huggingface.co/hw2942) to support replication and further research.

The remainder of this paper is organized as follows. Section 2 describes the data and task construction. Section 3 outlines the models, tokenization, fine-tuning procedures, and evaluation metrics. Section 4 presents the empirical results and financial applications. Section 5 concludes with key findings and directions for future research.

## 2 Data

### 2.1 Definition of Climate-Related Classification Tasks

The three climate-related classification tasks provide a framework for guiding our study from the initial stages of data collection, and it is essential to define them precisely. The first task is to determine whether a given text is related to climate issues. In our study, the scope of "climate relevance" is broad, encompassing corporate climate disclosures, climate policy, news on climate change, climate action, and climate risk. Texts that do not fall within this scope are classified as irrelevant to climate topics. Once a text is identified as climate-related, it is further categorized into either climate opportunity or climate risk, akin to sentiment analysis where the classification is between positive and negative. Text is labeled as climate opportunity if it primarily discusses business opportunities or the positive impacts of mitigating or adapting to climate change. Conversely, a text is categorized as climate risk if it focuses on business ventures or the negative impacts of climate change. For texts labeled as climate risk, a further distinction is made between physical risk and transition risk. Physical risks refer to the direct consequences of climate change on the environment and infrastructure, which can incorporate extreme weather events or gradual shifts in climate patterns. Transition risks stem from the shift towards a low-carbon economy and contain policy and legal risks, technology risks, market risks, and reputational risks.

### 2.2 Data Information

The raw data consist of CSR and ESG reports from A-share listed firms in China, supplemented by climate risk alerts from Wind. Financial data, involving monthly returns and firm characteristics, are obtained from CSMAR. We construct labeled datasets covering physical risk, transition risk, and climate opportunity. The physical risk dataset includes 500 sentences from corporate disclosures and climate risk alerts, identified using targeted keywords (e.g., extreme weather, drought, flood, wildfire, sea level rise). The transition risk dataset also contains 500 sentences, drawn from corporate reports and alerts related to energy transition and policy shifts. The climate opportunity dataset comprises 1000 sentences from CSR and ESG reports describing low-carbon technologies and strategies. Together, these yield 2000 climate-related sentences, of which 1500 (750 per category) are randomly selected for analysis. An additional 1500 climate-unrelated sentences are collected from the social and governance sections of ESG reports.

## 3 Models

### 3.1 ML Models

The ML models employed in this study include KNN, Naive Bayes, Logistic Regression, Decision Tree, Random Forest, and SVM. KNN [7,9] is a non-parametric method widely used in pattern recognition and data mining.

Naive Bayes, based on Bayes' theorem with an independence assumption, provides a simple yet effective approach to classification. Logistic Regression models the probability of categorical outcomes using explanatory variables. Decision Trees generate interpretable, tree-structured models that perform both prediction and feature selection. Random Forest [4,11,12], an ensemble of decision trees, improves accuracy and robustness against overfitting, making it useful in applications ranging from fraud detection to bioinformatics. Lastly, SVM [6,22] applies max-margin algorithms to identify optimal decision boundaries for classification and regression tasks.

Before training and testing the ML models, we apply preprocessing tailored to Chinese text. This includes removing punctuation, whitespace, and stop words, followed by word segmentation using jieba. The segmented tokens are transformed into numerical features via term frequencyinverse document frequency (TF-IDF) or bag-of-words (BoW). Model training and testing are implemented in sklearn, with configurations summarized in Table 1.

**Table 1.** ML Models Configuration

| ML models | Python package | Configuration |
|---|---|---|
| KNN | sklearn.neighbors.KNeighborsClassifier | Default |
| Naive Bayes | sklearn.naive_bayes.MultinomialNB | Default |
| Logistic Regression | sklearn.linear_model.LogisticRegression | solver='liblinear' |
| Decision Tree | sklearn.tree.DecisionTreeClassifier | Default |
| Random Forest | sklearn.ensemble.RandomForestClassifier | n_estimators=20 |
| SVM | sklearn.svm.LinearSVC | Default |

### 3.2 DL Models

LSTM, introduced by Hochreiter and Schmidhuber [13], is a recurrent neural network (RNN) architecture designed to address the vanishing gradient problem and capture long-term dependencies in sequential data. It consists of a memory cell and three gates—input, output, and forget—that regulate information flow. BiLSTM [10] extends this design by processing sequences in both directions, improving performance in NLP tasks. GRU [5], a simplified variant of LSTM, retains gating mechanisms but omits the output gate, offering comparable performance with fewer parameters and greater efficiency.

For DL training, we preprocess text by removing punctuation, whitespace, and stop words, followed by segmentation with jieba and tokenization via word embeddings. Models are implemented in TensorFlow.keras, with structural details provided in Table 2.

**Table 2.** DL Models Configuration

| DL Models | Structure configuration |
|---|---|
| LSTM1 | LSTM (100) + LSTM (50) + Dense (1) |
| LSTM2 | LSTM (50) + Dense (1) |
| BiLSTM | Bidirectional (LSTM (50)) + Dense (1) |
| GRU1 | GRU (100) + GRU (50) + Dense (1) |
| GRU2 | GRU (50) + Dense (1) |

### 3.3 LLMs

The three LLMs applied in this study—BERT-base-Chinese, Chinese-MacBERT-base, and RoBERTa—are derived from the BERT architecture. BERT [14], developed by Google, is a pre-trained bidirectional transformer encoder trained with masked language modeling (MLM) and next sentence prediction (NSP) objectives on large English corpora. It is then fine-tuned on labeled data for downstream NLP tasks.

BERT-base-Chinese adapts this framework to Chinese text, with 12 transformer layers, 12 attention heads, and 110 million parameters. MacBERT [8] improves upon BERT by introducing MLM as a correction task, replacing NSP with sentence order prediction, and training on extensive Chinese corpora, yielding strong performance across multiple Chinese NLP benchmarks. RoBERTa [17] further optimizes pre-training with dynamic masking, removal of NSP, longer sequences, extended training, and larger batch sizes, consistently outperforming BERT on major benchmarks.

In this study, these three pre-trained LLMs are fine-tuned on labeled datasets for climate-related classification tasks using the Hugging Face platform (Table 3).

**Table 3.** Information of pre-trained LLMs

| LLM pre-trained Models | Huggingface Link |
|---|---|
| BERT | bert-base-chinese |
| MacBERT | hfl/chinese-macbert-base |
| RoBERTa | hfl/chinese-roberta-wwm-ext |

### 3.4 Word Tokenizer

Effective tokenization is critical for NLP tasks, particularly in Chinese. This study applies different tokenization methods across ML, DL, and LLM models. For ML models, we compare TFIDF and BoW: TFIDF weights words by term frequency and inverse document frequency, while BoW represents documents as

word count vectors. Both yield identical results, so TFIDF is adopted as the standard tokenizer. For DL models, we employ Chinese Word Vectors [16] with 300-dimensional embeddings and a 50,000-word vocabulary, enabling semantic and syntactic representation of text. For LLMs, we use the BERT tokenizer based on the WordPiece algorithm, implemented via the transformers library:

$$BertTokenizer.from_pretrained(pretrained_model)$$

This tokenizer is applied for all three pre-trained LLMs (BERT-base-Chinese, MacBERT, and RoBERTa), with their Hugging Face links provided in Table 3.

### 3.5 Training and Fine-Tuning

As outlined in Sect. 2, the datasets contain 3000, 2000, and 1000 sentences for the three climate-related classification tasks, each balanced across two labels. For ML, DL, and LLM training, the data are randomly divided into equal training and testing subsets while preserving label balance: 1500/1500 for climate-related classification, 1000/1000 for climate risk versus opportunity, and 500/500 for physical versus transition risk. A validation set is incorporated during training and fine-tuning to optimize performance. This partitioning follows standard practice and ensures robust model evaluation. Details of the data partitioning scheme are provided in Table 4.

**Table 4.** The data partitioning scheme

| Classification Tasks/Models | ML Models | DL Models | LLMs |
|---|---|---|---|
| Climate related/unrelated | Train/Test: 1500/1500 | Train/Validate/Test: 1350/150/1500 | Train/Validate/Test: 1400/100/1500 |
| Climate risk/opportunity | Train/Test: 1000/1000 | Train/Validate/Test: 900/100/1000 | Train/Validate/Test: 900/100/1000 |
| Climate transition/physical risk | Train/Test: 500/500 | Train/Validate/Test: 450/50/500 | Train/Validate/Test: 450/50/500 |

### 3.6 Evaluation Metrics

This study focuses on three binary classification tasks, evaluating their performance using accuracy and F1-score as the primary metrics. These metrics are derived from the confusion matrix shown in Table 5, which provides a comprehensive breakdown of classification outcomes. To be specific, True Positive (TP) and True Negative (TN) represent instances where the predicted and actual labels correspond to the positive and negative categories, respectively. Conversely, False Positive (FP) and False Negative (FN) reflect cases of misclassification. The calculation of accuracy and F1-score is conducted using established formulas:

$$accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$F1 = \frac{2 * TP}{2 * TP + FP + FN}$$

**Table 5.** Confusion Matrix

| Actual/Predicted | Positive | Negative |
|---|---|---|
| Positive | True Positive (TP) | False Negative (FN) |
| Negative | False Positive (FP) | True Negative (TN) |

# 4 Empirical Results

## 4.1 Classification Results

To evaluate the robustness and reliability of experimental results, this study conducts ten iterations of training and fine-tuning processes across ML models, DL models, and pre-trained LLMs. On the climate-related classification task, as shown in Table 6, BERT-base-Chinese outperforms all models, achieving a maximum accuracy and F1 of 0.994, mean accuracy of 0.9897, and mean F1 of 0.9898. SVM leads all ML models, recording 0.976 across accuracy and F1 measures. In the DL category, GRU1 achieves the best performance, with accuracy (max = 0.9627, mean = 0.9549) and F1 scores (max = 0.9631, mean = 0.9555). Overall, LLMs exceed ML and DL models in all tasks. Notably, simpler ML models, including Naive Bayes, Logistic Regression, and SVM, surpass DL models despite the latter's advanced tokenization and architecture, suggesting the efficacy of traditional approaches in specific contexts.

**Table 6.** The results of climate-related classification task

| Models | | Max Accuracy | Mean Accuracy | Max F1 | Mean F1 |
|---|---|---|---|---|---|
| ML Models | KNN | 0.932 | 0.932 | 0.9327 | 0.9327 |
| | Naive Bayes | 0.9707 | 0.9707 | 0.9707 | 0.9707 |
| | Logistic Regression | 0.9733 | 0.9733 | 0.9732 | 0.9732 |
| | Decision Tree | 0.9367 | 0.9264 | 0.9361 | 0.9257 |
| | Random Forest | 0.9573 | 0.9507 | 0.9568 | 0.9498 |
| | SVM | 0.976 | 0.976 | 0.976 | 0.976 |
| DL Models | LSTM1 | 0.956 | 0.9465 | 0.9564 | 0.947 |
| | LSTM2 | 0.9533 | 0.9483 | 0.9531 | 0.9483 |
| | BiLSTM | 0.9533 | 0.9462 | 0.9538 | 0.947 |
| | GRU1 | 0.9627 | 0.9549 | 0.9631 | 0.9555 |
| | GRU2 | 0.962 | 0.9534 | 0.962 | 0.9541 |
| LLMs | BERT-base-Chinese | 0.994 | 0.9897 | 0.994 | 0.9898 |
| | Chinese-MacBERT-base | 0.993 | 0.9849 | 0.993 | 0.9848 |
| | RoBERTa | 0.992 | 0.988 | 0.992 | 0.988 |

**Table 7.** The results of climate risk and opportunity classification task

| Models | | Max Accuracy | Mean Accuracy | Max F1 | Mean F1 |
|---|---|---|---|---|---|
| ML Models | KNN | 0.934 | 0.934 | 0.936 | 0.936 |
| | Naive Bayes | 0.946 | 0.946 | 0.948 | 0.948 |
| | Logistic Regression | 0.966 | 0.966 | 0.966 | 0.966 |
| | Decision Tree | 0.868 | 0.8595 | 0.866 | 0.857 |
| | Random Forest | 0.952 | 0.934 | 0.952 | 0.934 |
| | SVM | 0.967 | 0.967 | 0.967 | 0.967 |
| DL Models | LSTM1 | 0.97 | 0.9637 | 0.97 | 0.9638 |
| | LSTM2 | 0.973 | 0.9612 | 0.973 | 0.96 |
| | BiLSTM | 0.969 | 0.961 | 0.9688 | 0.9609 |
| | GRU1 | 0.967 | 0.9637 | 0.9668 | 0.963 |
| | GRU2 | 0.97 | 0.9648 | 0.9699 | 0.9646 |
| LLMs | BERT-base-Chinese | 0.989 | 0.9855 | 0.989 | 0.985 |
| | Chinese-MacBERT-base | 0.993 | 0.989 | 0.993 | 0.989 |
| | RoBERTa | 0.99 | 0.9875 | 0.99 | 0.987 |

For the climate risk and opportunity task, detailed in Table 7, Chinese-MacBERT-base performs best, with accuracy and F1 both reaching a maximum of 0.993 and a mean of 0.989. SVM again dominates the ML models, yielding 0.967 in both accuracy and F1. For DL models, LSTM2 achieves the highest maximum accuracy and F1 score of 0.973, while GRU2 exhibits outstanding mean performance, with accuracy of 0.9648 and F1 score of 0.9646. These results reaffirm the superior performance of LLMs compared to ML and DL models. While SVM beats DL models in terms of mean accuracy and F1 scores, DL models display remarkable performance in achieving higher maximum values, highlighting the nuanced strengths of each approach.

Table 8 highlights the performance of various models on the climate physical and transition risk classification task. Chinese-MacBERT-base excels in maximum metrics with an accuracy and F1 score of 0.988, while RoBERTa demonstrates superior mean metrics (accuracy and F1 of 0.983). SVM attains the highest scores within ML models (maximum accuracy and F1 of 0.964 and 0.9636), whereas GRU1 outperforms other DL models with maximum values of 0.97 for both accuracy and F1. These findings emphasize the consistent dominance of LLMs, the competitive mean performance of ML models like SVM, and the maximum efficacy potential of DL models, revealing a nuanced hierarchy of model effectiveness.

From these findings, several insights can be drawn: First, LLMs persistently outperform ML and DL models for all climate-related tasks in both maximum and mean accuracy and F1 scores. For extracting meaningful insights from climate texts, LLMs should be leveraged for both English and Chinese, particularly with rapidly advancing tools like ChatGPT and DeepSeek. Second, as

**Table 8.** The results of climate physical and transition risk classification task

| Models | | Max Accuracy | Mean Accuracy | Max F1 | Mean F1 |
|---|---|---|---|---|---|
| ML Models | KNN | 0.926 | 0.926 | 0.924 | 0.924 |
| | Naive Bayes | 0.958 | 0.958 | 0.957 | 0.957 |
| | Logistic Regression | 0.958 | 0.958 | 0.957 | 0.957 |
| | Decision Tree | 0.91 | 0.8878 | 0.909 | 0.8845 |
| | Random Forest | 0.946 | 0.9334 | 0.9459 | 0.9325 |
| | SVM | 0.964 | 0.964 | 0.9636 | 0.9636 |
| DL Models | LSTM1 | 0.952 | 0.9438 | 0.9526 | 0.9439 |
| | LSTM2 | 0.95 | 0.9434 | 0.95 | 0.9427 |
| | BiLSTM | 0.948 | 0.9422 | 0.9476 | 0.9416 |
| | GRU1 | 0.97 | 0.9606 | 0.97 | 0.96 |
| | GRU2 | 0.966 | 0.9486 | 0.9657 | 0.948 |
| LLMs | BERT-base-Chinese | 0.982 | 0.9744 | 0.982 | 0.974 |
| | Chinese-MacBERT-base | 0.988 | 0.9798 | 0.988 | 0.9799 |
| | RoBERTa | 0.988 | 0.983 | 0.988 | 0.983 |

task complexity increases from the climate-related classification to the physical and transition risk tasks, the performance of all models decreases. Notably, the performance gap between SVM and LLMs is more pronounced in the physical and transition risk tasks than in the climate-related classification task. Consequently, for simpler climate classification tasks, ML models such as SVM are viable options, while LLMs offer superior performance for more complex tasks. Lastly, in cases where computational resources and time are limited, SVM is a suitable alternative for preliminary applications. However, with sufficient computational resources, LLMs should be explored for superior results in climate-related tasks, as the field evolves with the release of increasingly powerful models.

### 4.2 Financial Results

Building on the superior performance of LLMs, we apply the fine-tuned BERT-base-Chinese model to all ESG and CSR reports of A-share listed firms (20072024). Using three-level hierarchical classification, we extract sentence-level counting data on overall climate content, climate risk, climate opportunity, physical risk, and transition risk.

Firm-year climate metrics are then constructed as sentence proportions. The share of climate-related content is measured as climate-related sentences divided by total sentences in a report. Climate risk and opportunity proportions are calculated relative to climate-related sentences, while physical and transition risk proportions are calculated relative to climate-risk sentences.

To examine the stock market response to firms' climate parameters, we establish the regression model:

$$\text{Monthly Return}_{i,m,t} = \beta_0 + \beta_1 \text{Climate Parameter}_{i,t-1} + \beta_2 \text{Controls}_{i,t-1} + \gamma_t + \mu_j + \varepsilon_{i,m,t}$$

where $MonthlyReturn_{i,m,t}$ is the return of firm i in month m of year t, and $ClimateParameter_{i,t-1}$ is the calculated climate proportion parameters for firm i in year t-1. Table 9 presents the summary statistics of the variables used in the regression model. The model controls for a range of firm-specific characteristics, including monthly variables such as the logarithm of market capitalization, book-to-market ratio, momentum, and volatility, as well as annual variables such as leverage, capital expenditures to total assets (Capex/Total Assets), return on assets (ROA), and asset growth. Annual (monthly) variables are lagged by one year (month). Year and industry fixed effects are included, and standard errors are double-clustered at the firm and year levels.

**Table 9.** Summary statistics of variables in regression model

| Variables | Count | Mean | Std | Min | 25% | 50% | 75% | Max |
|---|---|---|---|---|---|---|---|---|
| Climate-related proportion | 147745 | 0.1397 | 0.0955 | 0.0000 | 0.0786 | 0.1231 | 0.1800 | 0.8699 |
| Climate risk proportion | 147745 | 0.0739 | 0.1097 | 0.0000 | 0.0000 | 0.0417 | 0.1094 | 1.0000 |
| Climate opportunity proportion | 147745 | 0.9124 | 0.1534 | 0.0000 | 0.8889 | 0.9545 | 1.0000 | 1.0000 |
| Physical risk proportion | 147745 | 0.2368 | 0.3508 | 0.0000 | 0.0000 | 0.0000 | 0.5000 | 1.0000 |
| Transition risk proportion | 147745 | 0.3655 | 0.4174 | 0.0000 | 0.0000 | 0.0000 | 0.7778 | 1.0000 |
| Monthly return | 147745 | 0.0081 | 0.1297 | –0.7246 | –0.0651 | –0.0038 | 0.0657 | 4.5631 |
| Ln(monthly market capitalization) | 147745 | 16.2819 | 1.1435 | 9.8627 | 15.4318 | 16.1611 | 16.9910 | 21.6668 |
| Book-to-market | 147745 | 0.6131 | 0.6416 | 0.0018 | 0.2830 | 0.4717 | 0.7701 | 27.1186 |
| Leverage | 147745 | 0.4608 | 0.1931 | 0.0080 | 0.3165 | 0.4693 | 0.6066 | 0.9967 |
| Capex/Total assets | 147745 | 0.0523 | 0.0474 | 0.0000 | 0.0185 | 0.0389 | 0.0716 | 0.5196 |
| ROA | 147745 | 0.0421 | 0.0665 | –0.9652 | 0.0154 | 0.0382 | 0.0699 | 0.6042 |
| Asset growth | 147745 | 0.2057 | 2.8677 | –0.6376 | 0.0153 | 0.0894 | 0.1998 | 312.3794 |
| Momentum | 147745 | 0.0070 | 0.0350 | –0.1617 | –0.0152 | 0.0028 | 0.0241 | 0.3595 |
| Volatility | 147745 | 0.1162 | 0.0582 | 0.0153 | 0.0787 | 0.1046 | 0.1398 | 1.3289 |

Table 10 reports regression results examining the relationship between five dimensions of climate exposure and monthly stock returns, controlling for year and industry fixed effects. The climate-related and physical risk proportion show no statistically significant association with returns, suggesting these exposures are not robustly priced in equity markets. In contrast, the coefficient on climate risk proportion turns negative and becomes statistically discernible once fixed effects are included, with an estimate of 0.0106, reaching the 1% threshold. Similarly, transition risk proportion yields negative values (-0.0029), with effects emerging at the 5% level. Climate opportunity proportion is positively associated with returns, with a magnitude of 0.0053 and attaining 5% significance in the fully specified model. Control variables perform consistently with

**Table 10.** The regression results relating monthly stock returns to climate parameters.

| | monthly return | | | | |
|---|---|---|---|---|---|
| climate-related proportion | –5.779e–05 | | | | |
| | (0.004) | | | | |
| climate risk proportion | | –0.0106*** | | | |
| | | (0.004) | | | |
| climate opportunity proportion | | | 0.0053** | | |
| | | | (0.002) | | |
| transition risk proportion | | | | –0.0029** | |
| | | | | (0.001) | |
| physical risk proportion | | | | | –0.0013 |
| | | | | | (0.001) |
| Ln(monthly market capitalization) | 0.0114*** | 0.0114*** | 0.0114*** | 0.0115*** | 0.0114*** |
| | (0.002) | (0.002) | (0.002) | (0.002) | (0.002) |
| book-to-market | –0.041*** | –0.041*** | –0.041*** | –0.041*** | –0.0409*** |
| | (0.006) | (0.006) | (0.006) | (0.006) | (0.006) |
| leverage | –0.007* | –0.007* | –0.0072* | –0.0068* | –0.0069* |
| | (0.004) | (0.004) | (0.004) | (0.004) | (0.004) |
| Capex/total assets | –0.0626*** | –0.0627*** | –0.0629*** | –0.0615*** | –0.0626*** |
| | (0.020) | (0.020) | (0.020) | (0.020) | (0.020) |
| ROA | –0.0885*** | –0.0892*** | –0.0889*** | –0.0891*** | –0.0883*** |
| | (0.018) | (0.018) | (0.018) | (0.018) | (0.018) |
| Asset growth | –0.0092*** | –0.0092*** | –0.0091*** | –0.0091*** | –0.0092*** |
| | (0.003) | (0.003) | (0.003) | (0.003) | (0.003) |
| momentum | –0.4961*** | –0.497*** | –0.4965*** | –0.497*** | –0.4962*** |
| | (0.103) | (0.103) | (0.103) | (0.103) | (0.103) |
| volatility | –0.0131 | –0.0131 | –0.0129 | –0.0132 | –0.0130 |
| | (0.048) | (0.048) | (0.048) | (0.048) | (0.048) |
| Year fixed effects | Yes | Yes | Yes | Yes | Yes |
| Industry fixed effects | Yes | Yes | Yes | Yes | Yes |
| N | 147745 | 147745 | 147745 | 147745 | 147745 |
| R-squared | 0.054 | 0.054 | 0.054 | 0.054 | 0.054 |
| Adj. R-squared | 0.053 | 0.053 | 0.053 | 0.053 | 0.053 |

*Notes:* Standard errors in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

asset pricing theory: firm size is positively and notably related to returns, while book-to-market ratio, Capex/total assets, ROA, asset growth, and momentum are substantially negative across specifications.

These findings imply that equity markets are beginning to differentiate among climate exposures—penalizing climate and transition risk while rewarding opportunity, but remain largely indifferent to physical and general climate fac-

tors. For policymakers and investors, this signals both progress in market-based climate risk integration and ongoing blind spots that merit further attention.

## 5 Conclusions

This paper introduces a novel study on Chinese climate-related text classification, focusing on dataset construction, model development, and practical implementation. We apply three climate-related classification tasks: detecting climate-related vs. unrelated content, climate risk vs. opportunity, and climate transition vs. physical risk. For these tasks, we collect and annotate datasets consisting of sentences from corporate reports of A-share listed companies in China, along with climate risk-related alert information from Wind. After training six ML models and five DL models, and fine-tuning three LLMs, we find that LLMs outperform all ML and DL models across all tasks, achieving the highest maximum and mean accuracy and F1 scores. However, SVM, an ML model, performs well in the simpler task of climate-related classification, offering a viable alternative when computing resources and training time are limited. To answer the question of whether markets price climate disclosure, this study employs the climate text classification approach based on the fine-tuned BERT model and regression analysis. Our findings demonstrate that Chinese capital markets penalize climate and transition risk exposure but reward climate opportunity, with limited evidence that overall climate disclosure volume and physical climate risk are priced into stock returns.

Future research can advance this study in four directions. First, expanding the dataset and incorporating additional sources (e.g., other media and information providers) would improve robustness. Second, model performance could be enhanced through systematic hyperparameter tuning rather than default configurations. Third, beyond the three core classification tasks examined here, more granular climate-related tasks should be investigated. Last, extending the framework to domains such as industry classification and regional segmentation would broaden its contribution to climatefinance research.

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